

Martin Luther University of Halle-Wittenberg
Department of Economics
Chair of Econometrics

Econometrics II
5. Time Series Models (Part 2)
Univariate Time Series Models

Christoph Wunder

Key questions and objectives

This chapter focuses on the following key questions and objectives:

- How can we model temporal patterns in time series data?
- How can we test and correct for unit roots?
- What is the best forecast for a univariate time series?

Outline

⑤ Univariate Time Series Models

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5.7.1 The first order autoregressive process: $AR(1)$ process

5.7.2 The first order moving average process: $MA(1)$ process

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5.7 Autoregressive processes

5.7.1 The first order autoregressive process: AR(1) process

- The **first order autoregressive process** (AR(1) process)

$$Y_t = \delta + \theta Y_{t-1} + \epsilon_t, \quad (5.16)$$

states that the current value of Y_t depends linearly on its previous value Y_{t-1} . We assume $|\theta| < 1$.

- The error term ϵ_t is assumed to follow a **white noise process**, i.e., the ϵ_t s are serially uncorrelated with zero mean and constant variance.
- Assuming that $E[Y_t]$ does not depend on t , the expected value of Y_t is

$$\mu \equiv E[Y_t] = \frac{\delta}{1-\theta}. \quad (5.17)$$

- Alternatively, we may define $y_t \equiv Y_t - \mu$ and write

$$y_t = \theta y_{t-1} + \epsilon_t. \quad (5.18)$$

- It holds that $Var(y_t) = Var(Y_t)$.
- Assuming stationarity, the variance of Y_t is:

$$Var(Y_t) = Var(\theta Y_{t-1} + \epsilon_t) = \theta^2 Var(Y_{t-1}) + Var(\epsilon_t) = \frac{\sigma^2}{1 - \theta^2}. \quad (5.19)$$

Note: assuming $|\theta| < 1$ is necessary to obtain a positive variance of Y_t .

- The joint distribution of all Y_t is characterized by the covariances between Y_t and its lags Y_{t-k} (**autocovariances**) for $k \geq 1$.
- Covariance between Y_t and Y_{t-1} is

$$\text{Cov}(Y_t, Y_{t-1}) = E[y_t y_{t-1}] = \theta \text{Var}(y_{t-1}) = \theta \frac{\sigma^2}{1 - \theta^2}. \quad (5.20)$$

- In general, the covariance between Y_t and Y_{t-k} is

$$\text{Cov}(Y_t, Y_{t-k}) = \theta^k \frac{\sigma^2}{1 - \theta^2}. \quad (5.21)$$

- Any two observations have a nonzero covariance (as long as $\theta \neq 0$).
- The covariance is smaller if the observations are further apart.
- The covariance depends only on k , not on t . This reflects stationarity.

5.7.2 The first order moving average process: MA(1) process

- The **first order moving average process** (MA(1) process)

$$Y_t = \mu + \epsilon_t + \alpha\epsilon_{t-1} \quad (5.22)$$

states that Y_t is, apart from μ , a weighted average of ϵ_t and ϵ_{t-1} .

- The **variances** are:

$$Var(Y_t) = (1 + \alpha^2)\sigma^2.$$

- The **covariances** are:

$$Cov(Y_t, Y_{t-1}) = \alpha\sigma^2$$

$$\text{Cov}(Y_t, Y_{t-k}) = 0, \text{ for } k = 2, 3, 4, \dots$$

- Observations that are two or more periods apart are uncorrelated.

5.7.3 General ARMA processes

- We define **general** autoregressive and moving average (ARMA) processes of order p and q , respectively. We assume ϵ_t to be a white noise process and $y_t = Y_t - \mu$ is the demeaned series.
- MA(q) process:

$$y_t = \epsilon_t + \alpha_1 \epsilon_{t-1} + \alpha_2 \epsilon_{t-2} + \dots + \alpha_g \epsilon_{t-g}. \quad (5.23)$$

- $AR(p)$ process:

$$y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \dots + \theta_p y_{t-p} + \epsilon_t. \quad (5.24)$$

- The ARMA(p, q) model combines both processes:

$$y_t = \theta_1 y_{t-1} + \dots + \theta_p y_{t-p} + \epsilon_t + \alpha_1 \epsilon_{t-1} + \dots + \alpha_q \epsilon_{t-q}. \quad (5.25)$$

- An AR model can be written as an MA model and vice versa.
- **Example:** AR(1) model written as MA(∞) model
 - AR(1) model:

$$y_t = \theta y_{t-1} + \epsilon_t. \quad (5.26)$$

- Substitute $y_{t-1} = \theta y_{t-2} + \epsilon_{t-1}$:

$$y_t = \theta^2 y_{t-2} + \epsilon_t + \theta \epsilon_{t-1}. \quad (5.27)$$

- Repeated substitution yields:

$$y_t = \theta^n y_{t-n} + \sum_{j=0}^{n-1} \theta^j \epsilon_{t-j}. \quad (5.28)$$

- For $n \rightarrow \infty$ and $|\theta| < 1$, we obtain an infinite moving average process:

$$y_t = \sum_{j=0}^{\infty} \theta^j \epsilon_{t-j}. \quad (5.29)$$

- The **lag operator** L (sometimes also denoted as **backshift operator** B) is defined as

$$L^p y_t = y_{t-p}. \quad (5.30)$$

- **Examples:**

$$Ly_t = y_{t-1},$$

$$L^{-1}y_t = y_{t+1},$$

$$L\mu = \mu \text{ (where } \mu \text{ is a constant),}$$

$$L^0 y_t \equiv 1.$$

- The AR(1) model can be written using the lag operator as follows:

$$(1 - \theta L)y_t = \epsilon_t. \quad (5.31)$$

That is, a combination of y_t and its lag, with weights 1 and $-\theta$, equals a white noise process.

- In general, the $AR(p)$ model can be written:

$$\theta(L)y_t = \epsilon_t, \quad (5.32)$$

where $\theta(L)$ is the **lag polynomial**

$$\theta(L) = 1 - \theta_1 L - \theta_2 L^2 - \dots - \theta_p L^p. \quad (5.33)$$

- The lag polynomial can be interpreted as a **filter**: if applied to a time series, it produces a new series. For example, if the filter $\theta(L)$ is applied to an $\text{AR}(p)$ process y_t , then it produces a white noise process ϵ_t .
- The **inverse** of $\theta(L)$, denoted $\theta^{-1}(L)$, is defined to satisfy

$$\theta^{-1}(L)\theta(L) = 1. \quad (5.34)$$

- If $\theta(L)$ is a finite order polynomial in L , then its inverse will be a polynomial of infinite order. For example, in the AR(1) case (with $|\theta| < 1$):

$$(1 - \theta L)^{-1} = \sum_{j=0}^{\infty} \theta^j L^j. \quad (5.35)$$

- The AR(1) model in MA(∞) form: multiplying the AR(1) model in 5.31 by 5.35 gives

$$(1 - \theta L)^{-1}(1 - \theta L)y_t = (1 - \theta L)^{-1}\epsilon_t,$$

$$y_t = \sum_{j=0}^{\infty} \theta^j L^j \epsilon_t,$$

$$y_t = \sum_{j=0}^{\infty} \theta^j \epsilon_{t-j}, \quad (5.36)$$

which is equivalent to the infinite moving average process in 5.29.

- The MA(q) model can be written using the lag operator as follows:

$$y_t = \alpha(L)\epsilon_t, \quad (5.37)$$

where the lag polynomial is

$$\alpha(L) = 1 + \alpha_1 L + \alpha_2 L^2 + \dots + \alpha_q L^q. \quad (5.38)$$

- If $\alpha(L)$ is invertible and $\alpha^{-1}(L)$ exists, we can also write

$$\alpha^{-1}(L)y_t = \epsilon_t. \quad (5.39)$$

- The inverse of the lag polynomial in the MA(1) case, $\alpha^{-1}(L)$, is (assuming $|\alpha| < 1$):

$$(1 + \alpha L)^{-1} = \sum_{j=0}^{\infty} (-\alpha)^j L^j. \quad (5.40)$$

- The MA(1) model can be written in AR(∞) form:

$$(1 + \alpha L)^{-1} y_t = \epsilon_t$$

$$\sum_{j=0}^{\infty} (-\alpha)^j y_{t-j} = \epsilon_t$$

$$(-\alpha)^0 y_t + \sum_{j=1}^{\infty} (-\alpha)^j y_{t-j} = \epsilon_t$$

$$y_t = - \sum_{j=1}^{\infty} (-\alpha)^j y_{t-j} + \epsilon_t$$

$$y_t = \alpha y_{t-1} - \alpha^2 y_{t-2} + \alpha^3 y_{t-3} - \dots + \epsilon_t$$

$$y_t = \alpha (y_{t-1} - \alpha y_{t-2} + \alpha^2 y_{t-3} - \dots) + \epsilon_t$$

$$y_t = \alpha \sum_{j=0}^{\infty} (-\alpha)^j y_{t-j-1} + \epsilon_t \quad (5.41)$$

- The **general ARMA model** can be written using lag polynomials as follows:

$$\theta(L)y_t = \alpha(L)\epsilon_t. \quad (5.42)$$

- The general ARMA model in $MA(\infty)$ representation is:

$$y_t = \theta^{-1}(L)\alpha(L)\epsilon_t. \quad (5.43)$$

- The general ARMA model in $AR(\infty)$ representation is:

$$\alpha^{-1}(L)\theta(L)y_t = \epsilon_t. \quad (5.44)$$

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- A stochastic process is called **strictly stationary** if its properties are unaffected by a change of time origin.
- A stochastic process is called **weakly stationary** if the means, variances, and covariances are independent of time. Formally:

$$\text{var}(Y_t) = E[(Y_t - \mu)^2] = \gamma_0 < \infty \quad (5.46)$$

$$\text{Cov}(Y_t, Y_{t-k}) = E[(Y_t - \mu)(Y_{t-k} - \mu)] = \gamma_k, k = 1, 2, 3, \dots \quad (5.47)$$

- Weak stationarity is also called **covariance stationarity**.
- A process is called **trend stationary** if it is covariance stationary apart from a perfectly predictable time trend.

- **Example:** the ARMA(2,1) process

$$y_t = 1.2y_{t-1} - 0.2y_{t-2} + \epsilon_t - 0.5\epsilon_{t-1} \quad (5.48)$$

is **nonstationary** because the sum of the autoregressive coefficients equals one (**unit root**). However, Δy_t is a **stationary** ARMA(1,1) process:

$$\Delta y_t = 0.2\Delta y_{t-1} + \epsilon_t - 0.5\epsilon_{t-1}.$$

- A series which becomes stationary after first differencing is said to be **integrated of order one**, denoted $I(1)$.
- If Δy_t is described by a stationary $ARMA(p, q)$ model, we say that y_t is described by an **autoregressive integrated moving average (ARIMA) model** of order $p, 1, q$.

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- The **autocorrelation function (ACF)** describes the correlation between Y_t and its lag Y_{t-k} as a function of k :

$$\rho_k = \frac{Cov(Y_t, Y_{t-k})}{Var(Y_t)} = \frac{\gamma_k}{\gamma_0}. \quad (5.49)$$

(Note that the **autocovariances**, γ_k , are dependent on the units in which the variables are measured. Hence, we standardize by defining **autocorrelations**.)

- The ACF characterizes the process by describing how long and how strongly a shock in the process (ϵ_t) affects the values of Y_t .

- For the **MA(1) model** with $Var(Y_t) = (1 + \alpha^2)\sigma^2$, $Cov(Y_t, Y_{t-1}) = \alpha\sigma^2$, and $Cov(Y_t, Y_{t-k}) = 0$ (for $k \geq 2$), we obtain:

$$\rho_1 = \frac{\alpha}{1 + \alpha^2}, \rho_2 = 0, \rho_3 = 0, \dots$$

- For the **MA(2)** model

$$y_t = \epsilon_t + \alpha_1 \epsilon_{t-1} + \alpha_2 \epsilon_{t-2},$$

we have

$$\begin{aligned} E[y_t^2] &= (1 + \alpha_1^2 + \alpha_2^2)\sigma^2, \\ E[y_t y_{t-1}] &= (\alpha_1 + \alpha_1 \alpha_2)\sigma^2, \\ E[y_t y_{t-2}] &= \alpha_2 \sigma^2, \\ E[y_t y_{t-k}] &= 0 \text{ (for } k = 3, 4, 5, \dots). \end{aligned}$$

Hence, the ACF is zero after two lags.

- **Note:** For an $MA(q)$ model the ACF is zero generally after q lags.

- The **sample autocorrelation function** gives the **estimated autocorrelation coefficients** as a function of k .
- The coefficient ρ_k can be estimated by

$$\hat{\rho}_k = \frac{\frac{1}{T-k} \sum_{t=k+1}^T (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\frac{1}{T} \sum_{t=1}^T (Y_t - \bar{Y})^2}, \quad (5.50)$$

where $\bar{Y} = (1/T) \sum_{t=1}^T Y_t$ is the sample average.

- In empirical application, it will usually not be the case that $\hat{\rho}_k$ is zero for an MA model of order $q < k$.
- We can use $\hat{\rho}_k$ to test the hypothesis that $\rho_k = 0$, i.e. testing MA($k - 1$) against MA(k) is done by testing $\rho_k = 0$.

- The **test statistic**

$$\sqrt{T} \frac{\hat{\rho}_k}{\sqrt{1 + 2\hat{\rho}_1^2 + \dots + 2\hat{\rho}_{k-1}^2}} \quad (5.51)$$

follows asymptotically a standard normal distribution under the null. So **critical values** are from the standard normal distribution.

- **Example:** testing $MA(0)$ versus the alternative $MA(1)$ is essentially a test of $\rho_1 = 0$. The test statistic is $\sqrt{T}\hat{\rho}_1$.
- The order of a **MA model** can be determined from an inspection of the sample ACF. It will give us a reasonable starting value for q .
- For **AR models**, the ACF is less helpful because autocorrelation coefficients do not cut off at a finite lag length.

- The k -th order sample partial autocorrelation coefficient, denoted $\hat{\theta}_{kk}$, is an estimate for θ_k in an AR(k) model.
- $\hat{\theta}_{kk}$ measures the additional correlation between Y_t and Y_{t-k} after adjustments have been made for the intermediate values $Y_{t-1}, \dots, Y_{t-k+1}$.
- **Example:** The estimated coefficient for Y_{t-2} in the AR(2) model

$$Y_t = \delta + \theta_1 Y_{t-1} + \theta_2 Y_{t-2} + \epsilon_t$$

gives $\hat{\theta}_{22}$.

- If the true model is an $AR(p)$ process, then estimating an $AR(k)$ model by OLS gives consistent estimators for the model parameters if $k \geq p$. In this case, we have

$$\text{plim } \hat{\theta}_{kk} = 0 \text{ if } k > p. \quad (5.52)$$

The autocorrelation function and partial autocorrelation function

- The partial autocorrelation coefficients can be used to determine the order of an **AR process**. For an $AR(p)$ model the partial autocorrelations will be close to zero after the p -th lag.
- Testing an $AR(k-1)$ model versus an $AR(k)$ model implies testing the null hypothesis that $\theta_{kk} = 0$.
- The **test statistic** $\sqrt{T}\hat{\theta}_{kk}$ is asymptotically standard normal. Hence, we reject θ_{kk} if $|\sqrt{T}\hat{\theta}_{kk}| > 1.96$.
- For **MA models**, the partial autocorrelations do not have a cut off point but tail off to zero (just like the autocorrelations in an AR model).

An AR(p) process is described by

- ① an ACF that is infinite in extent (it tails off).
- ② a PACF that is (close to) zero for lags larger than p .

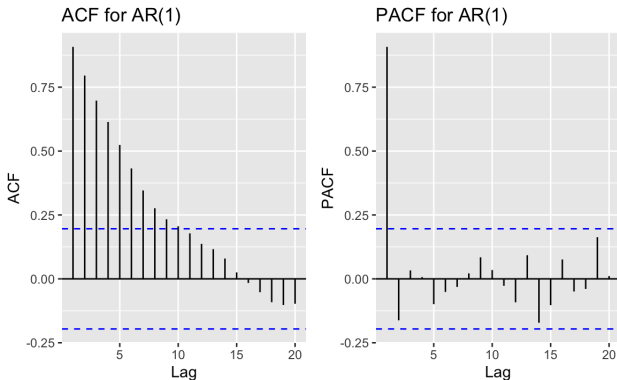
An MA(q) process is described by

- ① an ACF that is (close to) zero for lags larger than q .
- ② a PACF that is infinite in extent (it tails off)

- In other cases, an **ARMA model** may provide a parsimonious representation of the data.

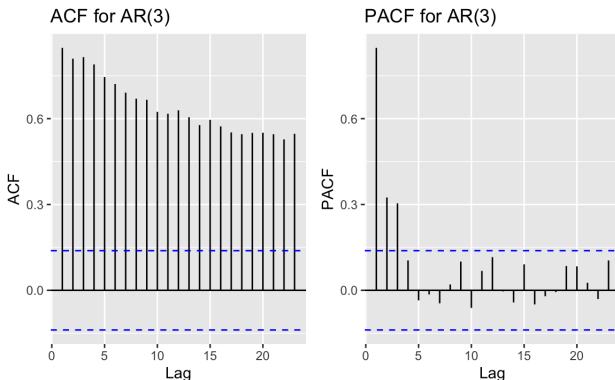
- **Example:** simulated AR(1) processes

```
library(forecast); library(ggplot2)
y <- arima.sim( model = list( ar = c(0.9)), n = 100 )
ggAcf( y ) + ggtitle("ACF for AR(1)")
ggPacf( y ) + ggtitle("PACF for AR(1)")
```



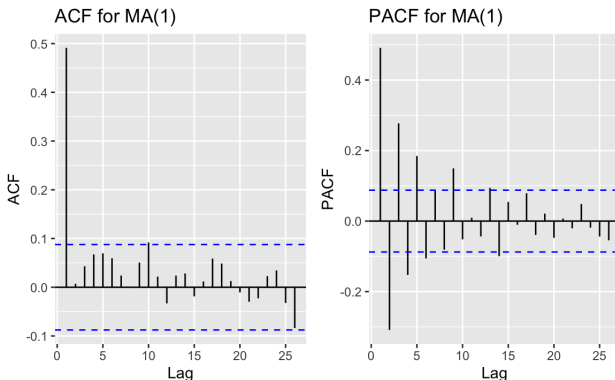
- **Example:** simulated AR(3) processes

```
y <- arima.sim( model = list( ar = c(0.4, 0.3, 0.2)), n = 200 )
```



- **Example:** simulated MA(1) processes

```
y <- arima.sim( model = list( ma = c(0.9)), n = 500 )
```



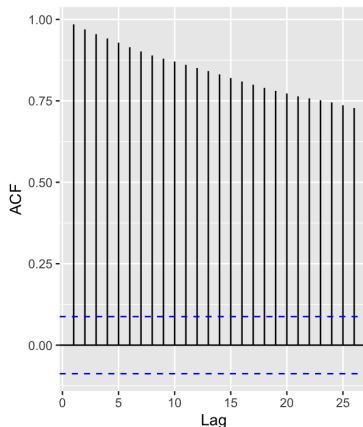
The autocorrelation function and partial autocorrelation function

- **Example:** $y_t = y_{t-1} + N(0, 1)$, **random walk (unit root)**, $I(1)$ series
 - $I(1)$ series wanders widely and has an **infinitely long memory**.
 - **Policy implication:** in models with unit roots, shocks (policy interventions) have persistent effects that last forever while in stationary models, shocks can only have a temporary effect.

Time series of unit root process



ACF for unit root



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5.10 Testing for unit roots

- Consider the AR(1) process:

$$Y_t = \delta + \theta Y_{t-1} + \epsilon_{t-1}. \quad (5.53)$$

- A test for a unit root considers the **null hypothesis** of $\theta = 1$.
- Test statistic:** we can use the standard t -statistic

$$\frac{\hat{\theta} - 1}{se(\hat{\theta})}, \quad (5.54)$$

where $\hat{\theta}$ is the OLS estimate for θ . (Note that OLS is consistent under the null.) $se(\hat{\theta})$ denotes the usual OLS standard error.

- Problem:** Under the null hypothesis the standard t -ratio does not have a t distribution, not even asymptotically.
- Solution:** critical values have to be taken from the **appropriate distribution**, provided by **Dickey and Fuller**.

- Critical values from the **Dickey-Fuller distribution**:

Sample size	Without trend		With trend	
	1%	5%	1%	5%
$T = 25$	-3.75	-3.00	-4.38	-3.60
$T = 50$	-3.58	-2.93	-4.15	-3.50
$T = 100$	-3.51	-2.89	-4.04	-3.45
$T = 250$	-3.46	-2.88	-3.99	-3.43
$T = 500$	-3.44	-2.87	-3.98	-3.42
$T = \infty$	-3.43	-2.86	-3.96	-3.41

Source: Verbeek (2017), Tab 8.1

- **Example:** using a 5% significance level in a one-tailed test of $H_0 : \theta = 1$ against $H_1 : |\theta| < 1$, the critical value in large samples is -2.86 (rather than -1.65 for the normal approximation).

- Usually, the regression model below is used to carry out the test:

$$\Delta Y_t = \delta + (\theta - 1)Y_{t-1} + \epsilon_t. \quad (5.55)$$

- Interpretation of a nonzero intercept term, $\delta \neq 0$:
 - In the stationary case ($|\theta| < 1$), δ reflects the nonzero mean of the series:

$$\delta = (1 - \theta)\mu \neq 0 \text{ if } \mu \neq 0 \text{ (for } |\theta| < 1\text{)}. \quad (5.56)$$

- In the unit root case ($\theta = 1$), δ reflects a **deterministic trend** in Y_t :

$$E[\Delta Y_t] = \delta$$

$$E[Y_t] = \delta + E[Y_{t-1}]$$

$$E[Y_t] = \delta + \delta + E[Y_{t-2}]$$

$$E[Y_t] = \delta t + Y_0,$$

where Y_0 is the given starting value.

- **Dickey–Fuller test with a trend:** testing for a random walk against the alternative of a trend stationary process:
 - Dickey–Fuller test with a trend is appropriate if a **graphical inspection** of the series indicates a clear positive or negative trend.
 - **Null hypothesis** of a random walk: $H_0 : \delta = \gamma = \theta - 1 = 0$.
 - **Test statistic:** it is common to use the t -ratio corresponding to $\theta - 1$, assuming that $\delta = \gamma = 0$. The regression model is

$$\Delta Y_t = \delta + (\theta - 1)Y_{t-1} + \gamma t + \epsilon_t. \quad (5.58)$$

- **Critical values** need to be adjusted for the fact that the testing regression includes a trend (i.e. we use critical values with trend from Dickey-Fuller distribution).
- If we reject the unit root hypothesis, we cannot conclude that the process for Y_t is likely to be stationary. Under the **alternative hypothesis** γ may be nonzero, i.e. the process may exhibit a linear deterministic trend. Hence, process for Y_t may be nonstationary (but trend stationary).

- **Augmented Dickey-Fuller (ADF) test:** the general strategy is that lagged differences, such as ΔY_{t-1} , ΔY_{t-2} , etc., are included in the regression.
- **Example:** in the AR(2) model

$$Y_t = \delta + \theta_1 Y_{t-1} + \theta_2 Y_{t-2} + \epsilon_t, \quad (5.59)$$

we have a unit root if $\theta_1 + \theta_2 = 1$. We can re-write the model as

$$\Delta Y_t = \delta + (\theta_1 + \theta_2 - 1)Y_{t-1} - \theta_2 \Delta Y_{t-1} + \epsilon_t \quad (5.60)$$

and use the OLS estimate of the coefficient for Y_{t-1} for testing the null hypothesis $\pi \equiv \theta_1 + \theta_2 - 1 = 0$. A time trend may be added to the regression.

- **General procedure:** Testing of a unit root in an $AR(p)$ process is based on the regression

$$\Delta Y_t = \delta + \pi Y_{t-1} + c_1 \Delta Y_{t-1} + c_2 \Delta Y_{t-2} + \dots + c_{p-1} \Delta Y_{t-p+1} + \epsilon_t, \quad (5.61)$$

where $\pi = \theta_1 + \dots + \theta_p - 1 = 0$ implies a unit root.

- The **test statistic** is the usual t -ratio.
- **Critical values** are the same as for the Dickey-Fuller test.
- The inclusion of the additional lags, in comparison to the standard Dickey-Fuller test, is done to make the error term in 5.61 asymptotically a white noise process, which is required for the distributional results to be valid.
- Because p is generally unknown, we should choose p fairly **large**. If too few lags are included the asymptotic distributions are not valid (because of autocorrelation in the residuals), and the tests may lead to seriously biased conclusions.

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Example: price/earnings ratio

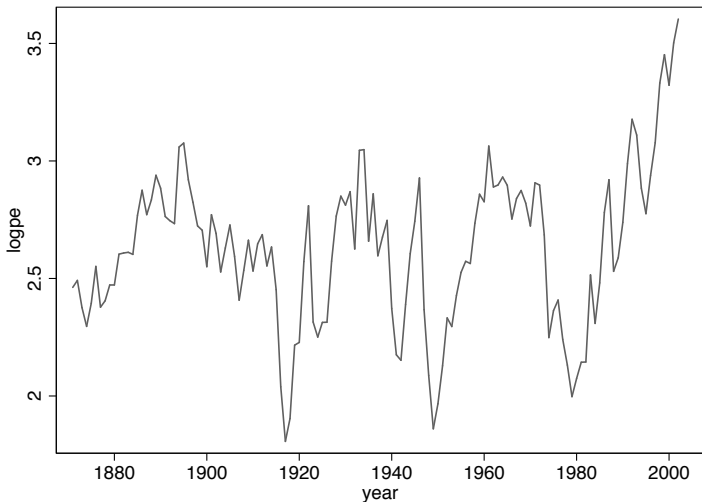
5.11 Example: price/earnings ratio

- **Data:** S&P Composite Stock Price Index and S&P Composite Earnings, here: price/earnings ratio over the period 1871–2002 ($T = 132$). (Dataset: `pe.dta`)

```
library(foreign)
d <- read.dta("pe.dta")
d.ts <- ts(d)
```

- **Question:** is the price/earnings ratio mean reverting?
- **Relevance** for forecasting future stock prices: high price/earnings ratios imply a decline in future stock prices to bring the ratio into line with its historical level.
- Graphical inspection of the data: it seems that the series fluctuates around a long-run average, although it sometimes takes many years for the series to revert to its mean.

- Annual log price/earnings ratio, 1871–2002:



Example: price/earnings ratio

- We estimate the Dickey-Fuller regression to test for unit roots in the log price/earnings ratio:

$$\Delta Y_t = \delta + (\theta - 1)Y_{t-1} + \epsilon_t. \quad (5.62)$$

```
> coeftest( dynlm( d(logpe) ~ 1 + L(logpe), data = d.ts ) )
t test of coefficients:

              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.334860   0.127902  2.6181  0.00990 **
L(logpe)     -0.124521   0.048471 -2.5690  0.01134 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The DF test statistic is -2.57 . The appropriate critical value at the 5% level is -2.89 . Hence, we cannot reject the null hypothesis of a first unit root.

Example: price/earnings ratio

- A range of **augmented Dickey-Fuller** tests should be carried out where additional lags of ΔY_t are included as RHS variables. This makes it more likely that the error term is white noise.
- **Example:** ADF(6) in R

```
> library(urca)
> ur.df(d$logpe, type = "drift", lags=6) # drift includes intercept

#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Value of test-statistic is: -1.8253 1.9856

Critical values for test statistics:
      1pct  5pct 10pct
tau2 -3.46 -2.88 -2.57
phi1  6.52  4.63  3.81
```

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Example: price/earnings ratio

- Overview of test results for various lag lengths (Verbeek 2017, Ch. 8.4):

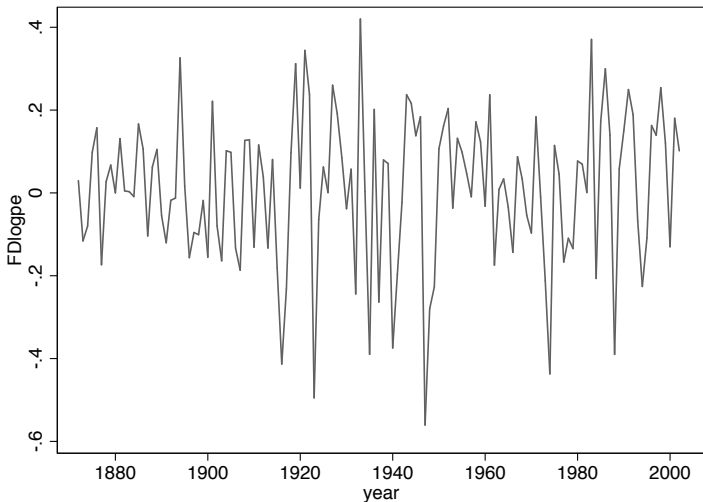
DF	$ADF(1)$	$ADF(2)$	$ADF(3)$	$ADF(4)$	$ADF(5)$	$ADF(6)$
-2.569	-3.000	-2.487	-2.503	-1.778	-1.627	-1.825

- Conclusion:** in general, the test results do not lead to a rejection of the null hypothesis that the series is nonstationary.
- First differenced series:** ADF tests clearly reject unit root.

DF	$ADF(1)$	$ADF(2)$	$ADF(3)$	$ADF(4)$	$ADF(5)$	$ADF(6)$
-10.588	-9.109	-7.310	-7.481	-6.431	-5.315	-4.210

- Note that the **change in the log** price/earnings ratio (i.e. the first differenced series) corresponds to the **growth rate** of the price/earnings ratio.

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5.7 Autoregressive processes

5.7.2 The first order moving average process: MA(1) process

5.7.3 General ARMA processes

5.8 Stationarity

5.10 Testing for unit roots

5.11 Example: price/earnings ratio

5.12 Estimating and choosing a model

5.13 Predicting with ARMA models

5.14 Example: forecasting short-term interest rates

5.12 Estimating and choosing a model

- ARMA models are usually estimated by **least squares** or **maximum likelihood** (for details, see Verbeek 2017, Ch. 8.6).
- Model **specification** is usually **data-driven**: estimated **autocorrelation** and **partial autocorrelation coefficients** give first idea.
- The model adequacy can be checked by a **residual analysis**: residuals of an adequate model should be approximately white noise. Hence, residual autocorrelations should be zero.
- The **Ljung-Box test** for **residual autocorrelation**:

$$Q_K = T(T+2) \sum_{k=1}^K \frac{1}{T-k} r_k^2 \sim \chi_{K-p-q}^2, \quad (5.63)$$

where the r_k are the estimated autocorrelation coefficients of the residuals, and $K > p + q$ is the number of lags being tested. The test has $K - p - q$ degrees of freedom.

Estimating and choosing a model

- **Model comparison:** all **model selection criteria** provide a trade-off between goodness-of-fit and the number of parameters used to obtain that fit.
- For example, if an MA(2) model would provide the same fit as an AR(10) model, we would prefer the first as it is more parsimonious.
- **Akaike's Information Criterion (AIC):**

$$AIC = \ln \hat{\sigma}^2 + 2 \frac{p + q + 1}{T}, \quad (5.64)$$

where $\hat{\sigma}^2$ is the estimated variance of ϵ_t .

- **Bayesian Information Criterion (BIC):**

$$BIC = \ln \hat{\sigma}^2 + \frac{p+q+1}{T} \ln T. \quad (5.65)$$

- Usually, “smaller is better”: the model with the smallest AIC or BIC value is preferred.

5.7 Autoregressive processes

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5.14 Example: forecasting short-term interest rates

- $$\mathcal{I}_T = \{Y_{-\infty}, \dots, Y_{T-1}, Y_T\}. \quad (5.66)$$

- $$Y_{T+h|T} \equiv E[Y_{T+h}|\mathcal{I}_T] \quad (5.67)$$

$$E[(Y_{T+h} - \hat{Y}_{T+h|T})^2 | \mathcal{I}_T]. \quad (5.68)$$

- The larger $\mathcal{I}_T$, the better the predictor will be. For example, $E[Y_{T+h}|Y_T, Y_{T-1}, Y_{T-2}, \dots]$ will usually be a better predictor than $E[Y_{T+h}|Y_T]$.
- To simplify notation, we consider forecasting of y_{T+h} (noting that $Y_{T+h|T} = y_{T+h|T} + \mu$).

How to determine the best predictor when y_t follows an **AR(p) process**?

- We assume that y_t follows an AR(1) process so that

$$y_{T+1} = \theta y_T + \epsilon_{T+1}. \quad (5.69)$$

- The best predictor for y_{T+1} at time T is

$$y_{T+1|T} = E[y_{T+1}|y_T, y_{T-1}, \dots] = \theta y_T + E[\epsilon_{T+1}|y_T, y_{T-1}, \dots] = \theta y_T. \quad (5.70)$$

- The best predictor for y_{T+2} at time T is

$$y_{T+2|T} = E[y_{T+2}|\mathcal{I}_T] = \theta E[y_{T+1}|\mathcal{I}_T] + E[\epsilon_{T+2}|\mathcal{I}_T] = \theta(\theta y_T) = \theta^2 y_T. \quad (5.71)$$

- In general, the best predictor for y_{T+h} at time T is

$$y_{T+h|T} = \theta^h y_T. \quad (5.72)$$

That is, the last value of y_T contains all information to determine the predictor for any future value.

- When $h \rightarrow \infty$, the predictor for y_{T+h} converges to zero (i.e. the unconditional expectation of y_t), provided that $|\theta| < 1$.
- The best predictor for Y_{T+h} with a nonzero mean ($\mu \neq 0$) is

$$\mu + y_{T+h|T} = \mu + \theta^h (Y_T - \mu). \quad (5.73)$$

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1000

- Predicting two-periods ahead gives:

$$E[y_{T+2}|\mathcal{I}_T] = E[\epsilon_{T+2}|\mathcal{I}_T] + \alpha E[\epsilon_{T+1}|\mathcal{I}_T] = 0, \quad (5.77)$$

because ϵ_{T+1} and ϵ_{T+2} are not in the information set.

- Thus, the MA(1) model is uninformative for forecasting two periods ahead. This also follows from the autocorrelation function of the process, because the ACF is zero after one lag. That is, the “memory” of the process is only one period.

How to determine the predictor when y_t follows an **ARMA(p, q) process**?

- For the general ARMA (p, q) model, the optimal predictors are:

$$\begin{aligned}
 y_{T+h|T} = & \theta_1 y_{T+h-1|T} + \dots + \theta_p y_{T+h-p|T} \\
 & + \epsilon_{T+h|T} + \alpha_1 \epsilon_{T+h-1|T} + \dots + \alpha_q \epsilon_{T+h-q|T},
 \end{aligned} \tag{5.78}$$

where

$$\begin{aligned}
 y_{T+k|T} &= y_{T+k} && \text{if } k \leq 0, \\
 \epsilon_{T+k|T} &= \epsilon_{T+k} && \text{if } k \leq 0, \\
 \epsilon_{T+k|T} &= 0 && \text{if } k > 0.
 \end{aligned}$$

5.14 Example: forecasting short-term interest rates

5.14 Example: forecasting short-term interest rates

- **Data:** time series of 1-month bond yields for the US. (Dataset: `irates.dta`)

```
library(foreign)
d <- read.dta("irates.dta")
d.ts <- ts(d)
```

- F-statistic: 2349 on 1 and 251 DF, p-value: $< 2.2e-16$

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Example: forecasting short-term interest rates

- **Dickey-Fuller tests of a unit root:**

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression drift

Call:
lm(formula = z.diff ~ z.lag.1 + 1)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.34902    0.15246   2.289  0.0229 *
z.lag.1     -0.04880    0.01963  -2.487  0.0135 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8215 on 251 degrees of freedom
Multiple R-squared:  0.02404,    Adjusted R-squared:  0.02016
F-statistic: 6.184 on 1 and 251 DF,  p-value: 0.01354

Value of test-statistic is: -2.4867 3.1029

Critical values for test statistics:
      1pct  5pct 10pct
tau2 -3.44 -2.87 -2.57
phi1  6.47  4.61  3.79
```

Example: forecasting short-term interest rates

- Dickey-Fuller tests suggest that a unit root in the short term interest rate cannot be rejected **statistically**. However, this does not mean that we have to accept the unit root hypothesis.
- **Economically**, it seems hard to defend nonstationarity of interest rates, although their persistence is known to be high. Even with stationarity it takes a very long time for the series to go back to its mean. So, we will continue with the assumption of stationarity.

Alternative time series models to describe short term interest rate

- Estimating AR(2) or ARMA(1,1) specifications, for example, does not result in a significant improvement.

Model	AIC	BIC
AR(1)	626.2633	636.8753
AR(2)	806.0216	816.6336
ARMA(1,1)	627.1394	641.2887

- **Example:** ARMA(1,1) in R:

```
> library(forecast)
> forecast::Arima(d$r1, order = c(1, 0, 1), include.constant = TRUE)
ARIMA(1,0,1) with non-zero mean
```

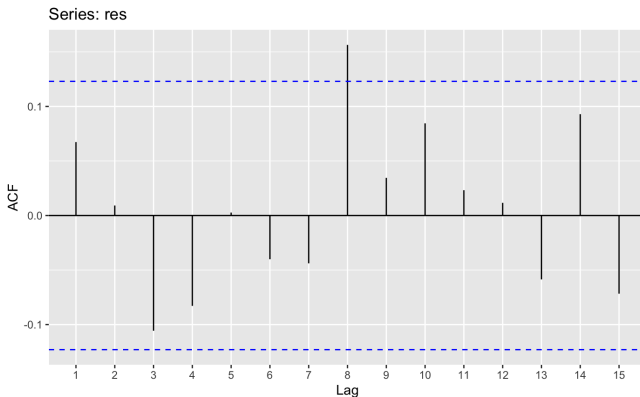
Coefficients:

	ar1	ma1	mean
	0.9411	0.0683	7.2314
s.e.	0.0211	0.0638	0.8749

```
sigma^2 estimated as 0.672:  log likelihood=-309.57
AIC=627.14   AICc=627.3   BIC=641.29
```

Testing for autocorrelation in the errors

- Autocorrelation function of the residuals of the AR(1) model:



- Only one of the residual autocorrelation coefficients is statistically significant. This is only weak evidence against the hypothesis that the error term in the AR(1) model is a white noise process.

- The **Ljung-Box test** does not reject the null hypothesis of a white noise process in the error term.

```
> fit <- forecast::Arima(d$r1, order = c(1, 0, 0), include.constant = TRUE)
> forecast::checkresiduals(fit, lag=12, plot = FALSE)
```

Ljung-Box test

```
data: Residuals from ARIMA(1,0,0) with non-zero mean
Q* = 15.656, df = 10, p-value = 0.1099
```

```
Model df: 2. Total lags used: 12
```

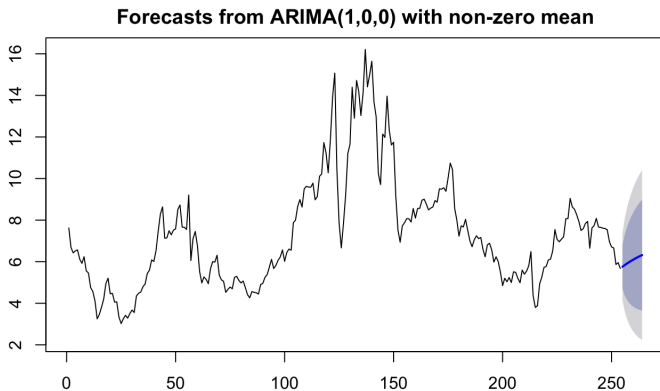
Forecasting

- Using $\theta = 0.95$ and the best predictor in equation 5.73, the optimal forecast of the short term interest rate 10 periods ahead is

$$Y_{T+10|T} = \mu + \theta^{10}(Y_T - \mu) = 7.15 + 0.951^{10}(5.677 - 7.15) = 6.26.$$

Example: forecasting short-term interest rates

```
pred <- forecast::forecast(fit, h = 10)
plot(pred)
```



Key terms

- ADF test
- Akaike's Criterion (AIC)
- AR process
- ARIMA process
- ARMA process
- autocorrelation function
- autocovariances
- backshift operator
- Bayesian Criterion (BIC)
- correlogram
- covariance stationarity
- Dickey-Fuller test
- drift
- lag operator
- lag polynomial
- Ljung-Box test
- MA process
- nonstationarity
- optimal predictor
- random walk
- stationarity
- trend stationarity
- unit root
- weak stationarity
- white noise process

References

This chapter is based on Verbeek (2017), Chapter 8.

Further readings: Greene (2011), Chapter 20.

Greene, W. H. (2011). *Econometric Analysis*. Prentice Hall International, Upper Saddle River, New Jersey.

Verbeek, M. (2017). *A Guide to Modern Econometrics*. John Wiley & Sons, Chichester.